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Final Examination, Business 33801 (AXP22): (Monday, 24 January 2022)

Read all questions carefully and write answers in the space provided. If you run out of space, you may write on the back of the exam pages; if you run out of space on the back of the exam, you may add additional pages of paper. Graphs must be thoroughly labeled for full credit. Show your work and explain all equations on numeric questions. Be concise – irrelevant remarks will not count for anything. Be careful to budget your time between various parts, taking into account the points allocated to each question (the number in parentheses); these points vary depending upon the question.

There are 14 questions with a total of 180 points possible. You have 180 minutes for the examination, so assign about one minute per point. Allocate your time wisely.

YOU MUST READ AND SIGN THE HONOR STATEMENT BELOW IN ORDER TO RECEIVE CREDIT FOR THE EXAMINATION.

I pledge my honor that I have not violated the Honor Code during this examination. I will not communicate any information related to this examination to *any* individual that has not already seen the examination for one week, **until after 31 January 2022**.

Sign your name here.

True-False-Uncertain (36 points total)

Indicate whether the italicized portion of each statement is True, False, or Uncertain and **EXPLAIN your answer**. You may take the non-italicized portion of the statement as true. No credit will be given for answers without explanation. *Read each question carefully.*

Question 1 [6 points] True, False, Uncertain? [Explain your answer.] Jay spends his money on two things - bottles of Scotch whisky (w) and games of golf (g). Each month, he consumes 12 of each. This month, however, the price of whisky has increased by £20 per bottle while the price of golf has fallen by £20 per game. His income available for purchasing whisky and golf is unchanged. *Jay is better off after these price changes.*

Question 2 [6 points] True, False, Uncertain? [Explain your answer.] Consider a competitive market. The short-run elasticity of supply is $\varepsilon_{sr}^s = 1.5$ and the long-run elasticity of supply is $\varepsilon_{lr}^s = 5$. The short-run elasticity of demand is $\varepsilon_{sr}^d = -2$ and the long-run elasticity of demand is $\varepsilon_{lr}^d = -4$. *If the government implements a unit tax, in both the short-run and the long-run, more than half of the tax will be absorbed by the sellers.*

Question 3 [6 points] True, False, Uncertain? [Explain your answer.] The government passes an operating tax in a constant-cost, competitive industry. The tax requires each firm pays a fixed annual fee of F dollars. Suppose that the introduction of the tax does not change the number of firms producing in the short run. *Total equilibrium output and price are unaffected in the short run, but the equilibrium price in the industry increases in the long run.*

Question 4 [6 points] True, False, Uncertain? [Explain your answer.] A direct price-discriminating monopolist (i.e., 3rd degree price discrimination) charges profit-maximizing prices p_A in market A and p_B in market B . The monopolist has a constant marginal cost of production that is the same for both markets. If at the profit-maximizing prices $p_A^* > p_B^*$, then *market A has a higher elasticity of demand (in absolute value) compared to market B.*

Question 5 [6 points] True, False, Uncertain? [Explain your answer.] In the competitive market for rechargeable lithium batteries, the long-run supply curve is upward sloping. *Some of the manufacturers of lithium batteries are earning economic profit in the long run.*

Question 6 [6 points] True, False, Uncertain? [Explain your answer.] Gloria's marginal utility of salsa is constant at $MU_s = 10$ and her marginal utility of wine is constant at $MU_w = 24$. The price of salsa is $p_s = 2$ and the price of wine is $p_w = 4$. *Gloria does not spend any of her income on salsa.*

Short questions (144 points)

Question 7 [28 points] Mitchell is a monopolist selling legal services in two market segments, A and B , with demands for legal services (in hours) given by the following functions:

$$q_a = 1000 - 10p_a,$$

$$q_b = 800 - 20p_b.$$

Mitchell's marginal and average opportunity costs are constant at $AC = MC = 20$.

(a). **[10 points]** Suppose that Mitchell can directly price discriminate across the markets and offer a different price per hour to each segment. What are the optimal quantities and prices for each segment?

(b). **[8 points]** Suppose that Mitchell is capacity constrained and can supply no more than 450 hours of legal services. That is, $MC = 20$ as before, but only for $q \leq 450$; Mitchell cannot produce greater output than 450. What is his optimal allocation of hours across the two market segments? [You may assume that Mitchell finds it optimal to work at capacity: $q_a^* + q_b^* = 450$.]

(c). **[8 points]** As in (b), suppose that Mitchell is capacity constrained at 450. Suppose, in addition, that the local government in market A has decided to regulate the price in segment A and impose a price ceiling of $\bar{p} = 25$. *Mitchell does not have to satisfy demand at that price*, but whatever hours he sells in market A cannot be sold for more than $\bar{p} = 25$ per hour. Importantly, there is no price regulation for market B .

Assuming that Mitchell chooses $p_a^* = 25$ in segment A and produces at capacity, $q_a^* + q_b^* = 450$, how much should he allocate to segment A and how much should he allocate to segment B ?

Question 8 [12 points] Claire runs a large firm which dominates an otherwise competitive market, giving her the ability to impact the price. Her marginal cost and average cost are both equal to 10. She faces a competitive fringe of price-taking firms that will inelastically supply exactly 100 units of output. The goods are homogenous, and the market demand function is $Q^d = 300 - 2p$.

How much should Claire sell in this market? What is the market price?

Question 9 [24 points] Consider the market for steel in the US. Steel demand in the US is given by the demand curve,

$$p = 100 - Q.$$

Steel supply to the US comes from domestic and foreign firms, $Q^s = q_{US}^s + q_f^s$. The US steel firms can supply steel at a constant unit cost of 20, per ton, and thus the US supply curve is a horizontal line at $p = 20$. The aggregate supply curve for foreign steel, however, is upward sloping and given by

$$q_f^s = p.$$

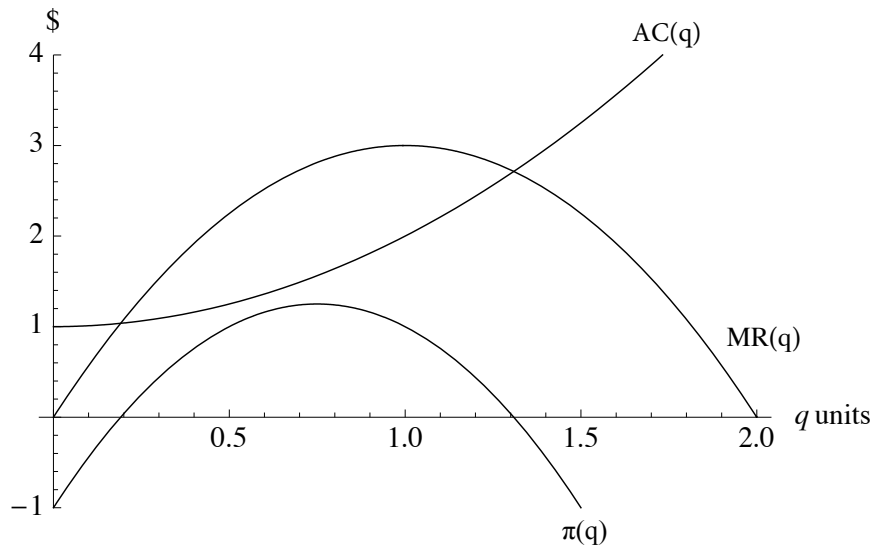
Note: As this is a competitive market, the equilibrium price and quantity of steel in the US market is found by equating demand to the sum of US and foreign supply.

(a). **[8 points]** What is the equilibrium price, p^* , for steel? How much is supplied by US firms, q_{us}^{s*} ? How much is supplied by foreign firms, q_f^{s*} ?

(b). **[10 points]** Suppose that the US government pays a subsidy to US steel producers equal to $s = 10$ per unit produced. What is the new equilibrium price paid by steel consumers in the US? How much steel is supplied by US firms and how much is supplied by foreign firms?

(c). [6 points] Explain in words the two sources of deadweight loss from the introduction of the steel subsidy.

Question 10 [6 points] Find one error in the following picture (there may be more than one). Briefly explain the error.



Question 11 [16 points] Phil runs a small, price-taking firm that has a variable (avoidable) cost function given by $C_{sr}^O(q) = q^2 + 10,000$, where q is monthly output.

(a). [6 points] At what price will Phil be indifferent to supplying the market and shutting down production in the short run? Explain.

(b). [10 points] If demand is perfectly inelastic with $Q^d = 15,000$, and if there are exactly 100 price taking firms (Phil's firm plus another 99) with the same cost function $C_{sr}^O(q) = q^2 + 10,000$, what is the short-run equilibrium price? Explain.

Question 12 [24 points] Suppose that the worldwide demand for cranberries, Q^d , is a function of the price of cranberries, p_c , and the price of apples, p_a .

$$Q_c^d = 1600 - 12p_c + 4p_a,$$

and the worldwide supply of cranberries is

$$Q_c^s = 200 + 4p_c - 20 \cdot temp,$$

where $temp$ is the average annual temperature of the cranberry growing regions of the world.

- (a). [4 points] Are apples and cranberries demand substitutes, demand complements or is it not possible to tell?
- (b). [6 points] Suppose that the price of apples is $p_a = 200$ and the average temperature is $temp = 30$. What is the equilibrium price and quantity of cranberries?
- (c). [6 points] What is the own-price elasticity of demand for cranberries at the equilibrium point in (b)? Is demand elastic or inelastic?
- (d). [8 points] If the worldwide average temperature increases next year but the price of apples remains constant at $p_a = 200$, describe what will happen to supply curve for cranberries? Using your answer to (c), what will the change in temperature do to the consumer expenditures on cranberries? [Hint: No further computations are needed to answer this question.]

Question 13 [14 points] A firm has three plants from which it can produce. Each plant has a capacity of 1000 units. The most efficient plant (plant A) has a total opportunity cost of production equal to $C_a(q_a) = q_a$ for producing $q_a \leq 1000$ units; plant B has a total opportunity cost of production equal to $C_b(q_b) = 2q_b$ for producing $q_b \leq 1000$ units; and the most costly plant has a total opportunity cost of production equal to $C_c(q_c) = 3q_c$ for producing $q_c \leq 1000$ units. Thus, the firm's total output across the three plants is $q = q_a + q_b + q_c \leq 3000$.

- (a). [6 points] Assume that the firm is a price taker and can sell as much as it likes at $p = 2.5$. How much should the firm produce at each plant?
- (b). [8 points] Assume again that the firm can sell as much as it likes at $p = 2.5$ on its domestic market, but the firm can also sell up to 1500 units on a foreign market at a higher price of $p = 4$ per unit. How much should the firm produce at each plant and how much should it sell in each market?

Question 14 [20 points] Consider a Cournot-output game in which two firms – Alex and Haley – operate as duopolists, simultaneously choosing what output to bring to market. The market demand curve is given by

$$p = 100 - \frac{1}{2}(q_a + q_h),$$

where q_a is the output Alex brings to market and q_h is the output that Haley brings to market.

- (a). [10 points] Suppose that Alex believes that Haley will bring $\hat{q}_h$ units to market. If Alex's marginal and average costs are constant at $MC_a = AC_a = 15$, how much output should Alex bring to market given her belief that Haley brings $\hat{q}_h$?
- (b). [10 points] Suppose that Haley's unit costs are higher than Alex's: $MC_h = AC_h = 20$. Confirm that $q_a^* = 60$ and $q_h^* = 50$ is a Nash equilibrium to this output game. [Hint: use your result in (a) to establish that this output for Alex is a best response to Haley's. Then verify, at the given outputs, that Haley's choice is also a best response to Alex's output.]

Suggested answers to exam questions

1 True. Jay can continue to purchase exactly what he had before with the same income (12 of each good). Given the relative prices have changed, however, he will likely be able to improve upon this bundle by substituting into golf and away from Scotch whisky because the the MRS will not equal to the new price ratio.

2 False. The sellers absorb more of the tax in the short run, but the reverse is true in the long run. In the short run, the supply elasticity is lower than the demand elasticity (in absolute value). Because the less elastic side of the market will absorb more of the tax, the sellers will absorb more of the tax. In the long-run, the reverse is true and the supply elasticity is larger than the demand elasticity (in absolute value) and thus the buyers will absorb more than half of the tax.

3 True. In the short run, the fee is not too large so as to cause firms to shut-down immediately, so the the short-run equilibrium price is unchanged because marginal cost is unchanged. In the long-run, because the minimum average cost must increase, some firms exit the industry and the equilibrium price rises to cover the higher average costs for the marginal (zero-profit) producing firm.

4 False. Suppose that the constant marginal cost of production is $MC = c$. The elasticity marginal-output rule (applied to each market segment) requires

$$\frac{p_A - c}{p_A} = \frac{1}{|\varepsilon_A|} \quad \text{and} \quad \frac{p_B - c}{p_B} = \frac{1}{|\varepsilon_B|}.$$

Because $p_A > p_B$,

$$\frac{p_A - c}{p_A} > \frac{p_B - c}{p_B}$$

and thus $|\varepsilon_A| < |\varepsilon_B|$. Market B is more elastic than market A at the optimal prices.

5 False/Uncertain. The entire profit from the market for rechargeable lithium batteries may be going to the producers of lithium rather than the manufacturers of the batteries. Where the profits go depends upon the relative scarcity of the various parts of the vertical production chain.

6 True. Gloria obtains $MU_s/p_s = 10/2 = 5$ utils per dollar spent on salsa, but $MU_w/p_w = 24/4 = 6$ utils per dollar spent on wine. Because marginal utilities are constant, spending a dollar on wine always generates more happiness than spending a dollar on salsa, so she should spend nothing on salsa.

7 (a). First we solve for the inverse demand functions with prices on the left hand side:

$$p_a = 100 - \frac{1}{10}q_a,$$

$$p_b = 40 - \frac{1}{20}q_b.$$

We can now compute the marginal revenues for each segment:

$$MR_a = 100 - \frac{1}{5}q_a,$$

$$MR_b = 40 - \frac{1}{10}q_b.$$

Setting these equal to $MC = 20$, yields

$$100 - \frac{1}{5}q_a = 20,$$

$$40 - \frac{1}{10}q_b = 20.$$

Solving each equation for output, we have $q_a^* = 400$ and $q_b^* = 200$. The optimal prices are found by substituting these outputs into the inverse demand functions:

$$p_a^* = 100 - \frac{1}{10}(400) = 60,$$

$$p_b^* = 40 - \frac{1}{20}(200) = 30.$$

(b). Given the answer in (a) where $q_a^* + q_b^* = 600 > 450$, we know that Mitchell will want to produce up to the full amount. Regardless, you were told to assume that this is the case. Mitchell will thus want to equate the marginal revenues between the two segments. This requires

$$100 - \frac{1}{5}q_a = 40 - \frac{1}{10}q_b.$$

Multiplying by 10, this simplifies to

$$q_b = 2q_a - 600.$$

In addition, $q_a + q_b = 450$ so it follows that

$$q_b = 2(450 - q_b) - 600,$$

or $q_b = (900 - 600)/3 = 100$. Consequently, $q_a^* = 450 - 100 = 350$.

(c). The key in this part is to note that the firm's MR in segment A is $MR_a = 25$ for all output up to $q_a = 700$, which is greater than capacity. Thus, as in (b), the solution is to set the marginal revenues equal to each other:

$$25 = 40 - \frac{1}{10}q_b.$$

We conclude that $q_b^* = 150$. Thus, $q_a^* = 300$.

8 First, we compute the large-firm's residual demand function:

$$q^d(p) = (300 - 2p) - 100 = 200 - 2p.$$

With this demand function, we then compute the optimal output and price. The firm's residual demand curve is $p = 100 - \frac{1}{2}q$. Using the marginal output rule, we have

$$MC = 10 = MR(q) = 100 - q,$$

so $q^* = 90$. The associated price is $p^* = 100 - \frac{1}{2}(90) = 55$.

9 (a). The equilibrium price will be $p^* = 20$. At this price, demand is $Q^* = 100 - 20 = 80$. Supply by foreign firms is $q_f^{s*} = p = 20$, and thus the supply by domestic firms is $q_{us}^{s*} = 80 - 20 = 60$.

(b). Because supply is horizontal around the intersection of demand and supply, the entire subsidy will be passed through to the buyers in terms of a lower price. Hence, $p_b^* = p_1^* = 20 - 10 = 10$. At this price, foreign supply will be $q_f^{s*} = p = 10$. Demand will be $Q^d = 100 - p = 90$, and so US firms will supply the residual of $q_{us}^{s*} = 90 - 10 = 80$.

(c). There are two sources of deadweight loss. First, the subsidy encourages excessive steel production/consumption. Output increase from the efficient level of 80 to the inefficient level of 90. The marginal buyer of steel at $p^* = 10$ values the steel at 10, but it costs US firms 20 to produce it.

Second, the subsidy leads to the wrong firms producing the steel. There are 10 units of steel that could be produced by foreign firms at a cost less than 20, the cost of the US firms.

10 There is one basic inconsistency that could be described in several different ways. The MR and AC curves are measured in \$/unit and not simply \$ as the vertical axis suggests; profit $\pi(q)$, however, is measured in total \$. Related, marginal revenue is either constant (with price taking firms) or it is downward sloping (with monopoly), but in the figure it is first increasing which is not possible for a marginal revenue function. Indeed, it looks like the AC and MR curves are really meant to be total cost and total revenue curves, respectively, given that these curves intersect at outputs where profit is zero.

11 (a). We need to find the minimum $AC_{sr}^O(q)$. This minimum occurs where $MC = AC$, so in the present case,

$$AC_{sr}^O(q) = q + \frac{10,000}{q} = MC_{sr}(q) = 2q.$$

Thus, the curves cross at $q = \sqrt{10,000} = 100$. At $q = 100$,

$$AC_{sr}^O(100) = MC_{sr}(100) = 200.$$

Thus, the firm produces zero for any price below $p = 200$ and at $p = 200$, the firm is indifferent between producing $q = 0$ and $q = 100$.

(b). Because demand is inelastic and equal to $Q^d = 15,000$, and because there are 100 firms, we know that each firm must produce 150 units in equilibrium. For this to be the case, it must be that $p^* = MC_{sr}(150) = 2(150) = 300$. Because this exceeds the critical price of 200 from part (a), it is a short-run equilibrium price.

12 (a). Apples and cranberries are demand substitutes because the demand for cranberries is increasing in the price of apples.

(b). Setting supply equal to demand, we have

$$1600 - 12p_c + 4(200) = 200 + 4p_c - 20(30),$$

so $p_c^* = 175$. The corresponding $Q_c^* = 200 + 4(175) - 600 = 300$.

(c). Elasticity of demand is

$$\varepsilon^d = \frac{p_c}{Q_c} \frac{dQ^d}{dp_c} = \frac{175}{300}(-12) = -7.$$

Demand is elastic.

(d). The demand curve remains fixed but the supply curve shifts inward as temperatures rise (i.e., less is supplied at any given price with the temperature increases). Because the before and after equilibrium points lie on the demand curve, and given that demand is elastic from (b), it follows that the reduction in output will lower consumer expenditures on cranberries. Explain.

13 Note that the firm will use plant A for the first 1000 units, plant B once plant A is at capacity, and plant C once plant B is at capacity. Thus, the firm's marginal cost curve is

$$MC(q) = \begin{cases} 1 & q < 1000 \\ 2 & 1000 < q < 2000 \\ 3 & 2000 < q < 3000. \end{cases}$$

(a). A price-taking firm's MR is equal to the price, $p = 2.5$. For $q < 1000$, $MC = 1 < 2.5$, so the firm should increase output. For $1000 < q < 2000$, $MC = 2 < 2.5$, and so the firm should continue to increase output. For $2000 < q < 3000$, $MC = 3 > 2.5$, so the firm should reduce output. The optimal output is $q^* = 2000$ with plants A and B operating at capacity and plant C idle.

(b). The firm's marginal revenue for the first 1500 units is $MR = 4$, because those units would first be sold on the foreign market. For $q > 1500$, the firm's MR falls to $MR = 2.5$, as in part (a). Notice that at $q = 1500$, $MR = 4 > MC$, so those 1500 units should be produced and sent to the foreign market. For additional output $1500 < q < 2000$, $MR = 2.5 > MC = 2$, and so the firm will want to produce at least 2000 for the domestic market. For $q > 2000$, $MR = 2.5$, but $MC = 3$, so the firm should not produce above $q^* = 2000$. As in (a), the firm produces $q^* = 2000$ across plants A and B. The firm sells 1500 units to the foreign market and 500 units to the domestic market.

14 (a). Alex's residual demand curve is simply

$$p = 100 - \frac{1}{2}q_a - \frac{1}{2}\hat{q}_h = \left(100 - \frac{1}{2}\hat{q}_h\right) - \frac{q_a}{2}.$$

Alex's corresponding marginal revenue (using the trick of "same intercept, twice the slope") is

$$MR_a = \left(100 - \frac{1}{2}\hat{q}_h\right) - q_a.$$

Thus, the marginal output rule requires

$$\left(100 - \frac{1}{2}\hat{q}_h\right) - q_a = 15,$$

or $q_a^* = 85 - \frac{1}{2}\hat{q}_h$.

(b). Plugging $q_h^* = 50$ into Alex's best-response function in (a), we have

$$q_a^* = 85 - \frac{1}{2}(50) = 60,$$

as required. Now consider Hayley's best response to $q_a^* = 60$. Her residual demand becomes

$$p = 100 - \frac{1}{2}(60 + q_h) = 70 - \frac{1}{2}q_h.$$

Thus, marginal revenue is

$$MR_h = 70 - q_h.$$

Setting this equal to $MC_h = 20$, we have $q_h^* = 50$, as required.